

PH 525P: Electronics Laboratory Practicum

Course Code	PH 525P
Course Name	Electronics Laboratory Practicum
L-T-P-C	0-0-5-3
Prerequisites	Electronics (PH 514)
Intended for	Core for M.Sc. and I-Ph.D. in Physics
Mutual Exclusion	NA
Approval	10th Senate (Jun 2016)
Revised	64th BoA (May 2026) - syllabus revised

Preamble

This laboratory course is designed to provide students with hands-on experience in the design, analysis, and implementation of fundamental electronic circuits and systems commonly used in physics and engineering applications. The course introduces practical aspects of analog and digital electronics through experiments. Students will gain experience in constructing and testing circuits such as transistor amplifiers, logic gates, filters, converters, multiplexers, counters, and multivibrators, and will learn the principles of interfacing sensors with microcontroller platforms such as Arduino. Through laboratory exercises and a mini-project, students will develop practical skills in circuit design, measurement, troubleshooting, and instrumentation, enabling them to apply electronic principles to real-life scientific and technological problems.

Course Modules

No

Laboratory Modules

1. To design and use bipolar junction transistors (BJT) as an amplifier and switch, based on common emitter (CE), common collector (CC), and common base (CB) configurations. Verify and draw the IV characteristics.
2. Seven-segment display. Output verification of Logic Gates.
3. Design of Integrator, Differentiator, low pass, and high pass filter using an OpAmp IC 741.
4. Design and verify A/D and D/A converters using OpAmp.
5. Design of 4-bit Multiplexer and Demultiplexer using flip flops.
6. Design of 4-bit Shift registers and Counters using flip flops.
7. Design of Astable and Monostable Multivibrator using IC 555.
8. Development and testing of an Arduino-based sensor circuit: Arduino-Based Sensor Interfacing, Firmware Development, Circuit Design, and Calibration
9. Design of Wein Bridge and Colpitts oscillator.

Project: Developing the working model of an electronic instrument for real-life application.

Text Books

1. Millman and Halkias, *Integrated electronics*: McGraw-Hill, (2001).
2. A. P. Malvino and D. P. Bates *Electronic Principles*: (7th Edn) McGraw-Hill (2006).

References

1. D. P. Leach, A. P. Malvino and G. Saha, *Digital Principles and Applications*: (6th Edn), Tata McGraw Hill (2007).
2. A. K. Maini, *Digital Electronics-Principles, Devices and Applications*: John Wiley & Sons (2007).
3. Sedra and Smith, *Microelectronic circuits*, Oxford publications, sixth edition 2013.